

$$\begin{aligned}
 &\Rightarrow D(AB) = C^{-1} \text{ (Theorem 2(e))} \Rightarrow (D(AB))B^{-1} = C^{-1}B^{-1} \\
 &\Rightarrow (DAB)B^{-1} = C^{-1}B^{-1} \text{ (Theorem 2(a))} \Rightarrow (DA)(BB^{-1}) = C^{-1}B^{-1} \text{ (Theorem 2(a))} \\
 &\Rightarrow (DA)I = C^{-1}B^{-1} \text{ (By Defn. of inverse)} \\
 &\Rightarrow DA = C^{-1}B^{-1} \text{ (Theorem 2(e))} \Rightarrow (DA)A^{-1} = C^{-1}B^{-1}A^{-1} \\
 &\Rightarrow D(AA^{-1}) = C^{-1}B^{-1}A^{-1} \text{ (Theorem 2(a))} \Rightarrow DI = C^{-1}B^{-1}A^{-1} \text{ (By defn. of inverse)} \\
 &\Rightarrow \boxed{D = C^{-1}B^{-1}A^{-1} \text{ (Theorem 2(e))}} \quad \square
 \end{aligned}$$

16) Suppose A and B are $n \times n$ matrices, B is invertible, and AB is invertible, $n \times n$. Show that A is invertible.

Solution: By defn. of invertible matrix, there exists a unique $n \times n$ matrix B^{-1} such that $BB^{-1} = B^{-1}B = I$.

By defn. of invertible matrix, there exists a unique $n \times n$ matrix C such that $(AB)C = C(AB) = I$. $C = (AB)^{-1}$

$$\begin{aligned}
 C(AB) &= I \Rightarrow (C(AB))B^{-1} = IB^{-1} \Rightarrow ((CA)B)B^{-1} = IB^{-1} \text{ (Theorem 2(a))} \\
 &\Rightarrow (CA)(BB^{-1}) = IB^{-1} \text{ (Theorem 2(a))} \Rightarrow (CA)I = IB^{-1} \text{ (By defn. of inverse)} \\
 &\Rightarrow CA = B^{-1} \text{ (Theorem 2(e))}. \text{ By Theorem 6(a), we know that } C \\
 &\text{ is invertible. } \therefore \text{ By Defn, there exists a unique matrix } C^{-1} \text{ such} \\
 &\text{ that } C^{-1}C = CC^{-1} = I \\
 &C^{-1}(CA) = C^{-1}B^{-1} \Rightarrow (C^{-1}C)A = C^{-1}B^{-1} \text{ (Theorem 2(a))} \\
 &\Rightarrow IA = C^{-1}B^{-1} \text{ (By Defn. of inverse)} \Rightarrow \boxed{A = C^{-1}B^{-1} \text{ (Theorem 2(e))}}
 \end{aligned}$$

By Theorem 6(b), BC is invertible and $(BC)^{-1} = C^{-1}B^{-1}$.
 $\therefore A = (BC)^{-1}$. By Theorem 6(a), $(BC)^{-1}$ is invertible, and $((BC)^{-1})^{-1} = BC = A^{-1}$.

Claim: BC is a solution of the equation $AX = I$ and $XA = I$ where X is an $n \times n$ matrix

Proof: $A(BC) = (C^{-1}B^{-1})(BC) = (C^{-1}B^{-1}B)C \text{ (Theorem 2(a))}$
 $= (C^{-1}(B^{-1}B))C \text{ (Theorem 2(a))} = (C^{-1}I)C \text{ (By defn. of inverse)}$
 $= C^{-1}C \text{ (Theorem 2(e))} = I \text{ (By defn. of inverse)}$